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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 541 — Past Questions & Materials
by Renance

Course: EMT 541
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 500

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Paper 1 of 1 — EMT 541 METHODS IN ENVIRONMENTAL ANALYSIS 111 2021,2022

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

2021/2022 ACADEMIC SESSION SECOND SEMESTER

EMT 541: Methods in Environmental Analysis III (3 Units)

Instruction: Answer 4 questions, at least one from each section

Time Allocated: 2 hours 30 Minutes

Section A

1. (a) Discussed the two techniques employed in solvent extraction (7 marks).
- (b) Discussed the principle of ion-exchange chromatography (10.5 marks)
2. (a) Discussed the technique for column chromatography (11 marks)
- (b) Explain the mobile phase in gas chromatography? (7.5 marks)
3. (a) Define Gas Chromatography (2 marks)
- (b) Explain the following (i) injection port (ii) stationary phase (iii)

Solid support

- (iv) Oven (10 marks)
- (C) What are the major differences among ion-exchange, column and gas chromatography (7.5 marks).

Section B

4. (a) Explain the term solvent extraction (8 $\frac{1}{2}$ marks)
- (b) why is solvent extraction important for chromatography (9marks)
5. (a) Briefly explain the following (17 $\frac{1}{2}$ marks)
- (i) Chromatography
- (ii) Extract

(iii) Extractant

(iv) Adsorbent

(v) Solvent

6. (a) If you were given a sample containing a mixture of water and crude oil, kindly tell us how you can separate the water out from the crude oil. (10marks)

(b) Differentiate between solute and solvent and give examples. (7 $\frac{1}{2}$ marks)

Section C

7. (a) List the methods of classifying activation methods

(b) What is the most important advantage of neutron activation analysis?

8. (a) What is the difference between radiochemical method and Isotope Dilution Analysis (IDA)?

(b) List the most important of radiometric titration

quantity radioactive substance ~~that~~ of the known ^{spectra} that is added to us